GURUMURUGAN A

Backend Software Engineer

Coimbatore, India | gurumurugan3004@gmail.com | +91 9025775674 | linkedin.com/in/gurumurugana | github.com/gurumurugan123

PROFESSIONAL SUMMARY

Backend Software Engineer with 1+ year of production experience building Django REST microservices on PostgreSQL, MongoDB, Redis, and Kafka. Cut API response times **~25%** and improved SQL query performance **~20%** through database and caching optimization. Built distributed backend systems independently (a Celery-style async job queue, a double-entry ledger engine) and mentor junior engineers on Django and microservices practices within a production team.

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Backend:** Django, Django REST Framework, Celery, REST APIs, gRPC
- **Databases:** PostgreSQL, MongoDB
- **Caching & Messaging:** Redis, Apache Kafka
- **Authentication:** Keycloak, JWT
- **DevOps & Tools:** Docker, Kubernetes, Jenkins, GitHub Actions, Git, Azure (dev/test environments)
- **Testing:** API testing with Postman

WORK EXPERIENCE

Python Backend Developer — Impiger Technologies *Aug 2025 – Present*

- Design and maintain Django REST microservices for an enterprise land-valuation platform running in production.
- Optimized PostgreSQL queries and Redis caching strategies, cutting API response time by **~25%** and improving SQL query performance by **~20%**.
- Implemented Keycloak-based JWT authentication and structured, traceable error logging across services.
- Built REST APIs for workflow automation, document handling, and cross-service data access using Django REST Framework.
- Containerized services with Docker and deployed them via Kubernetes and Jenkins/GitHub Actions CI/CD pipelines across dev, staging, and production.
- Set up and maintained dev and test environments on Azure; validated API functionality and edge cases with Postman before release.
- **Mentor 2 junior developers** on Django REST Framework conventions, code review practices, and microservices debugging.

Software Developer Intern — Impiger Technologies *Feb 2025 – Jul 2025*

- Developed REST APIs in Django within a microservices architecture; improved query execution time **~10–15%** through indexing and query optimization.
- Worked with PostgreSQL, Redis, and Docker in a production backend environment.

PROJECTS

LAVMIS — Land Valuation Management Information System

Python | Django | PostgreSQL | MongoDB | Redis | Kafka | Keycloak | Celery | Docker | Kubernetes

- Contributed to a Kubernetes-deployed microservices platform with API gateway routing and Kafka-based cross-service synchronization.
- Built a **Prediction microservice** with a config-driven Random Forest training pipeline (scikit-learn) and Celery-based batch retraining for property-value inference with confidence scoring.

TaskFlow — Background Job Processing Platform

Django | Redis | PostgreSQL | Docker | github.com/gurumurugan123/TaskFlow

- Built a **Celery-style asynchronous job queue from scratch** using Redis and BRPOPLPUSH-based workers.
- Implemented idempotency keys, exponential backoff retries, and dead-letter queues for reliable job processing.

LedgerVault — Digital Wallet & Double-Entry Ledger Engine

Python | Django | PostgreSQL | Redis | JWT | Docker | github.com/gurumurugan123/LedgerVault

- Built a fintech-style wallet backend on double-entry accounting principles with **ACID-compliant transfers** using SELECT FOR UPDATE row locking.
- Added idempotency keys, Redis-based rate limiting, JWT authentication, and a GitHub Actions CI pipeline.

EDUCATION

B.E. in Computer Science and Engineering *2021 – 2025*

Rathinam Technical Campus | CGPA: 7.6